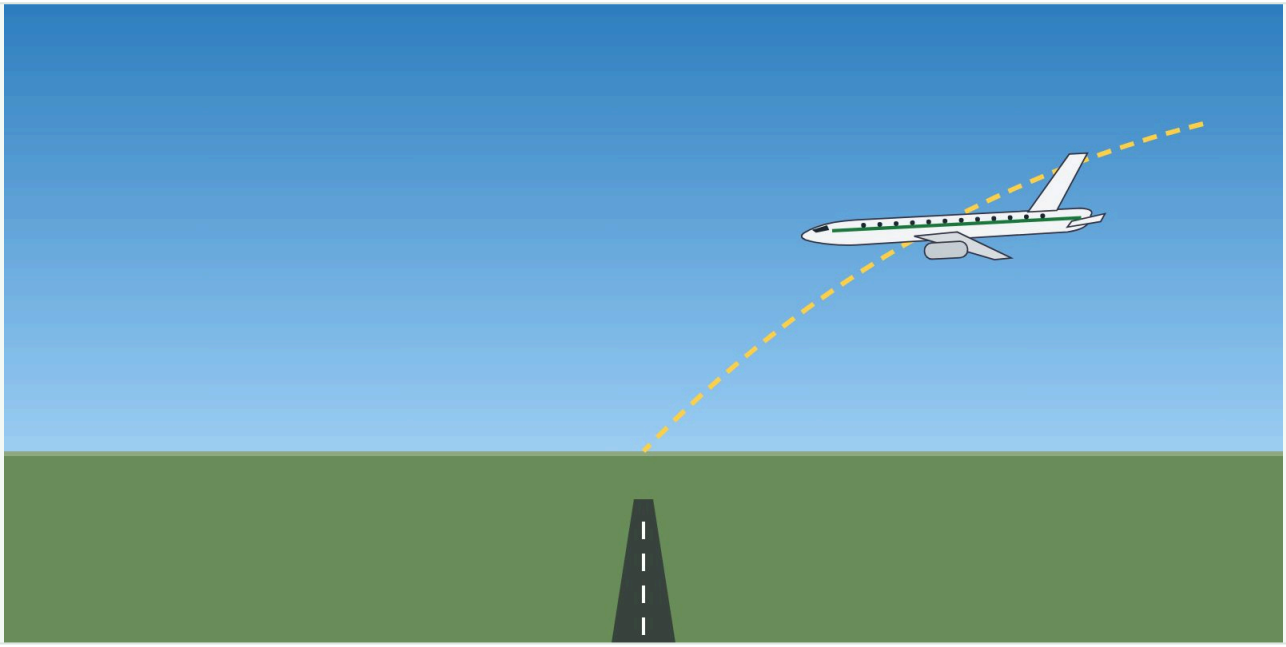
MODULE 2 · PBN OPERATIONS

Lesson 04: RNP Approaches: LNAV, LNAV/VNAV, LPV

Lesson 4 of 5 · about 8 min

LEARNING OBJECTIVES

- Distinguish the RNP APCH minima lines.
- Apply the temperature limits for baro-VNAV.
- Fly RNP approaches with the correct technique and go-around criteria.



LNAV, LNAV/VNAV or LPV: fly the minima you are authorised for.

Minima lines on an RNP chart

Minima	Guidance	Type
LNAV	Lateral only — flown with CDFA to an MDA	2D
LNAV/VNAV	Lateral + barometric vertical guidance (baro-VNAV) to a DA	3D
LP	Lateral SBAS guidance to an MDA	2D
LPV	Lateral + vertical SBAS guidance to a DA	3D

Use only the minima line your aircraft is certified and the company is authorised for. In EASA rules, **RNP APCH** (LNAV, LNAV/VNAV, LP, LPV) does not need a specific approval, but crews must be trained; **RNP AR APCH** requires a specific approval (SPA.PBN).

Temperature and baro-VNAV

- In cold air, the baro-VNAV path is **lower and shallower** than published; in hot air it is higher and steeper.

- Charts with LNAV/VNAV minima state a **temperature limit** — below it, LNAV/VNAV must not be used (use LNAV minima or another approach).
- Check the aerodrome temperature against the chart before every LNAV/VNAV approach.

Technique

- Brief the approach with the chart and the FMS: minima, vertical path angle, temperature limit, missed approach.
- Be stabilised by the company gates; use the autopilot and the appropriate vertical mode as your SOP requires.
- Monitor lateral and vertical deviations. Go around if the deviation limits in your SOP are exceeded, or on any navigation alert (e.g. loss of integrity) — unless the required visual references are established.
- The missed approach is part of the RNP procedure — fly it in LNAV as published.

KEY TAKEAWAYS

- LNAV and LP = 2D to an MDA; LNAV/VNAV and LPV = 3D to a DA.
- RNP APCH: no specific EASA approval but training required; RNP AR: specific approval.
- Respect the chart temperature limit for baro-VNAV.
- Go around on excessive deviation or navigation alert.

KNOWLEDGE CHECK

The OAT is below the minimum temperature published for LNAV/VNAV on the chart. What do you do?

A. Fly LNAV/VNAV anyway

B. Do not use LNAV/VNAV minima — use LNAV or another approach

C. Increase the DA by 50 ft

D. Fly faster

Answer: B. Below the temperature limit the baro-VNAV path is not protected.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.